

```
$ psql -c "SELECT ... FROM students JOIN courses ON ..."
```

SQL JOIN

// 各種 SQL JOIN

INNER

□□

LEFT

□□□□

RIGHT

□□□□

FULL

□□

CROSS

□□□□

```
$ cat query-plan.md
```



01 / NOT DESCRIBED NOT DESCRIBED JOIN

□□□□□□□□□□□□□□

02 / NOT DESCRIBED NOT DESCRIBED

□□□□□□□

03 / NOT DESCRIBED NOT DESCRIBED NOT DESCRIBED

ON vs WHERE □□□

04 / NOT DESCRIBED NOT DESCRIBED

□□□□□□□□□□

□□□□□□□□□□□□□□□□□□□□ = LEFT□□□ = INNER□

PART 01

NO GLYPH NO GLYPH JOIN

□□□ ID: 1,2,3,4 × □□□ ID: 2,3,5

INNER JOIN

NO GRAPH



```
1  SELECT
2      s.student_id,
3      s.name AS student_name,
4      c.course_name
5  FROM students AS s
6  INNER JOIN courses AS c
7      ON s.student_id = c.student_id;
```



RESULT SET

学生 ID 1, 2, 3, 4

学生 ID 2, 3, 5

→ 学生 ID 2, 3

x 学生 ID 1, 4

x 学生 ID 5

LEFT JOIN

HH
NO GRAPH

□□□□□□

```
1  SELECT
2      s.name,
3      c.course_name,
4      CASE
5          WHEN c.course_name IS NULL
6          THEN '□□□' ELSE '□□□'
7      END AS status
8  FROM students AS s
9  LEFT JOIN courses AS c
10     ON s.student_id = c.student_id;
```

RESULT SET

→ □□□ID 1, 2, 3, 4 □□□□

→ 1 □ 4 □□□□□□ NULL

□□



LEFT JOIN □□□□□

1

```
1 SELECT c.customer_name
2 FROM customers c
3 LEFT JOIN orders o
4   ON c.customer_id = o.customer_id
5 WHERE o.order_id IS NULL;
6 -- □□□□□□
```

2

```
1 SELECT s.name,
2        COUNT(c.course_id) AS course_count
3 FROM students s
4 LEFT JOIN courses c
5   ON s.student_id = c.student_id
6 GROUP BY s.student_id, s.name;
7 -- □□□□□□ count = 0
```

LEFT JOIN + IS NULL □□□□□□□□□□—INNER JOIN □□□□

RIGHT □ FULL OUTER □ CROSS

RIGHT JOIN

□□□□□□□□□□

A LEFT JOIN B ≡ B RIGHT JOIN

A □  LEFT □□□

FULL OUTER JOIN

 NULL□

MySQL  LEFT ∪ RIGHT

□ UNION □□□

CROSS JOIN

 3 × 4  = 12 
 × 

```
1  -- MySQL □□ FULL OUTER JOIN
2  SELECT * FROM students s LEFT JOIN courses c ON s.student_id = c.student_id
3  UNION
4  SELECT * FROM students s RIGHT JOIN courses c ON s.student_id = c.student_id;
```

PART 02

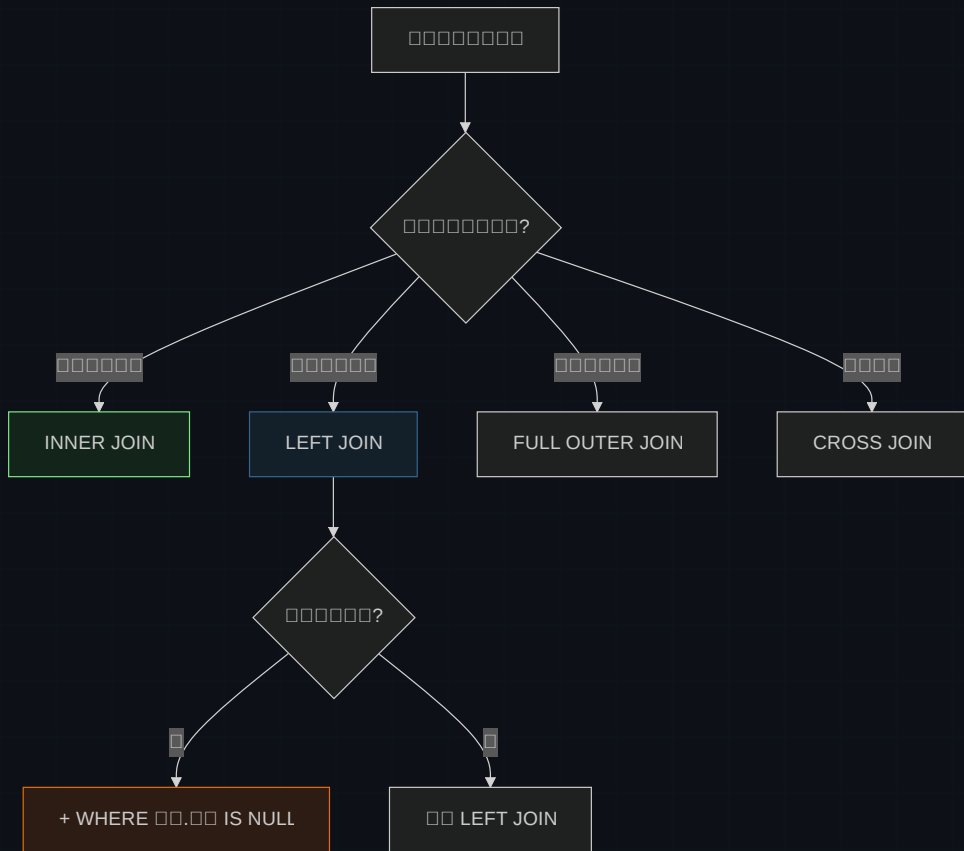
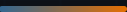


$$\square\square\square + \square\square\square\square\square$$

JOIN

JOIN	Table 1	Table 2	Result	Performance
INNER JOIN	10000	10000	1000	★★★★★
LEFT JOIN	100	10000	10000 NULL	★★★★★
RIGHT JOIN	10000	100	100 NULL	★★
FULL OUTER	100	100	10000 NULL	★★
CROSS JOIN	100	100	1000000	★

\$ 90% INNER JOIN LEFT — 10000000



PART 03

□□□□□ON vs WHERE

90% □ LEFT JOIN bug □□□

CHECKPOINT_01

LEFT JOIN courses c ... WHERE c.course_name = 'Math' □□□□

A □□□□□□□□□□ Math □□□ NULL

B □□□□ Math □□□—LEFT JOIN □ WHERE □□ INNER □□

C □□□□□LEFT JOIN □□□□□ WHERE



```
1  -- ❌ WHERE JOIN 表名 → NULL 表名 INNER 表
2  SELECT * FROM students s
3  LEFT JOIN courses c ON s.student_id = c.student_id
4  WHERE c.course_name = 'Math';    -- 表名 Math 表
5
6  -- ✅ ON JOIN 表名 → 表名
7  SELECT * FROM students s
8  LEFT JOIN courses c
9      ON s.student_id = c.student_id
10     AND c.course_name = 'Math';  -- 表 Math 表 NULL
11
12 -- ✅ 表名 → IS NULL 表 WHERE
13 SELECT * FROM students s
14 LEFT JOIN courses c ON s.student_id = c.student_id
15 WHERE c.student_id IS NULL;
```

ON = JOIN 表名 WHERE = JOIN 表名

PART 04





```
1  -- 查询所有学生的选课记录
2  -- students ← enrollments → courses
3
4  SELECT
5      s.name AS student_name,
6      c.course_name,
7      e.enrollment_date
8  FROM students s
9  INNER JOIN enrollments e ON s.student_id = e.student_id
10 INNER JOIN courses c ON e.course_id = c.course_id;
```

数据库

UNIQUE KEY (student_id, course_id) 数据库

数据库 JOIN

数据库 INNER JOIN 数据库



JOIN □□□

1 ON

```
1 -- ✖ □□□□
2 SELECT * FROM students s
3 INNER JOIN courses c;
4 -- □□□ = □□□ × □□□
```

2

```
1 -- ✔ □□□□□□□□
2 CREATE INDEX idx_courses_student_id
3 ON courses(student_id);
```

DEBUG FLOW

1. EXPLAIN □□□□□ → 2. □□ ON □□ → 3. □□□□□□□□ → 4. JOIN □□ 5 □□□□□□□□

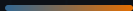


DO

- ✓ `students AS s`
- ✓ `SELECT *`
- ✓ `LEFT JOIN` `RIGHT JOIN`

DON'T

- × `ON` →
- × `LEFT` `INNER` →
- × `WHERE` → `LEFT` `INNER`



INNER = □□

□□□□□□□□

LEFT = □□

NOT NULL
NOT NULL
NOT NULL
NOT NULL + IS NULL □□□□□

ON vs WHERE

JOIN NOT NULL
NOT NULL
NOT NULL
NOT NULL vs JOIN □□□

□□

NOT NULL
NOT NULL
NOT NULL
NOT NULL
NOT NULL
NOT NULL
NOT NULL
NOT NULL ON □□

□□□□□□□□□□□□□□□□ — □ = LEFT□□□ = INNER□

Tables joined.

□□□□□□□□□□ JOIN □□□

```
$ EXPLAIN SELECT ... FROM a LEFT JOIN b ON ...  
inner → left → on/where → index
```